# Deliberation Costs: LLM and VLM Negotiators for Head-On Robot Encounters in Single-Lane Corridors

Roshan George, Martin Glavin, Edward Jones, and Brian Deegan

***Abstract*—A companion study benchmarked classical negotiation policies for the head-on corridor encounter, where two autonomous mobile robots meet in a single lane and one must reverse into a refuge. Here we ask whether foundation-model agents can take that role, evaluating nine large language and vision-language models as negotiators under the same physics and encounter seeds. In a centralized configuration with planner delay estimates in the prompt, reasoning-enabled models choose the right yielder in 92–93% of 200 encounters and reach $-1.0$ to $-1.7$ s mean excess delay relative to the best forced yielder, because they sometimes pick a better refuge than the planner default. Disabling the reasoning trace makes the same models worse than a hand-written rulebook (+11 to +30 s versus +2.9 s). Charging inference latency as time both robots stand still adds about 3.4 s of excess per second of latency, so the reasoning models beat cost bidding only with sub-second answers; no measured configuration combined such latency with reasoning-level quality. Decentralized agents with egocentric views fail mainly by disagreement, in model-specific ways: one model has both robots yield on 84% of encounters, two others have both proceed on 29–31%. Replacing the rulebook fallback with a fair random tiebreak resolves every conflict but lowers right-yielder rates from 63–71% to 48–63%, whether or not the agents see the draw. Code and per-episode results are released.**

***Index Terms*—Large language models, vision-language models, multi-robot systems, negotiation, mechanism design, autonomous mobile robots.**

## I. Introduction

WHEN two autonomous mobile robots meet head-on in a corridor too narrow to pass, someone must reverse into a pull-in bay or out of the lane entirely. A companion benchmark [1] compared reactive, fixed-priority, rule-based, reservation, cost-bidding, group-bidding, Clarke-pivot, and learned negotiators under identical physics and communication models, and found that cost bidding with accurate costs tracks the best forced yielder — a hindsight reference that tries both yield decisions — while degrading gracefully under bid noise and misreporting. That benchmark deliberately excluded one agent family: foundation models that read a textual or visual description of the encounter and decide in natural language.

Language-model negotiators are attractive for a specific reason: the corridor decision is a small, well-posed optimization — pick the yielder minimizing total delay — yet it is exactly the kind of social, two-party choice that language models are claimed to handle. Warehouses are increasingly populated by robots from multiple vendors with no shared planner; an agent that can reason over a broadcast state and pick a sensible yielder without bespoke protocol engineering would be valuable. The question is whether current models can, and what it costs.

We attach a large-language-model (LLM) and vision-language-model (VLM) policy adapter to the companion benchmark and evaluate nine models (local 12–49B-parameter weights on a single Grace-Blackwell node plus a hosted 120B endpoint) across arms that isolate decision quality, inference cost, input modality, and coordination structure. All arms draw encounters from the same seed sequence, so arms run at equal size share encounters one to one. Our contributions:

1) a prompt-and-parse LLM/VLM negotiator for the corridor encounter, with centralized (*mediator*) and decentralized (*agents*) modes, text and image inputs, JSON-schema constraint, response caching, and latency charged into the simulator as decision time;
2) a model screen in which reasoning-enabled models score below the forced-yielder reference through refuge selection, and an ablation in which this advantage disappears when the reasoning trace is disabled;
3) a deployed-cost analysis: charging measured latency as frozen decision time adds about 3.4 s of excess per second of latency, so the reasoning models' quality margin survives only for sub-second answers;
4) an egocentric decentralized study in which disagreement, not decision quality, is the dominant failure, with model-specific modes (both-yield vs. both-proceed);
5) a protocol-default experiment: a fair random tiebreak in place of the rulebook fallback resolves every conflict but lowers decision quality, because the rulebook names the right yielder on 66–71% of conflicted encounters and the draw on 40–51%; showing the draw to the agents changes one model's stated reasoning but not the outcome.

## II. Related Work

Language models have been applied to negotiation [2], [3], multi-agent coordination games, and robot task planning; [4] uses a foundation model for high-level deadlock resolution. These studies evaluate agents in bespoke environments with bespoke metrics; none compares foundation-model negotiators against a classical optimal baseline under identical

All authors are with the Connaught Automotive Research (CAR) group and the Ryan Institute, University of Galway, Ireland (e-mail: brian.deegan@universityofgalway.ie).

DRAFT, not for distribution.

physics, which the companion benchmark [1] provides. Vision-language models are increasingly used for robot perception and instruction following, but we are not aware of a controlled comparison of text, image, and egocentric-image inputs for an inter-robot *decision* task. Closest in structure, Chen et al. [5] compare centralized and decentralized LLM planners for multi-robot collaboration, the axis our mediator and agents modes span.

The classical alternative is market-based coordination, which allocates tasks and resources by bidding [6], [7]; multi-agent path finding has also been cast as a combinatorial auction [8]. The cost-bidding baseline we compare against belongs to this family. Negotiated path planning for non-cooperative robots [9] and opinion-driven decision making for narrow-corridor navigation [10] address related conflicts without language models.

Our tiebreak arms connect to game theory: public randomization resolves anti-coordination games as a correlated equilibrium [11], and in bargaining theory the disagreement point shapes negotiated outcomes [12]. We test whether stating the default changes what language-model agents choose. Coin flipping between mutually distrusting parties requires a commitment protocol [13]; our draw is a simpler publicly computable hash that assumes non-adversarial robots.

## III. Benchmark and Agents

### A. The corridor encounter

We use the companion benchmark [1] unchanged. Two TurtleBot3-Waffle-class unicycle robots (0.26 m/s forward, 0.15 m/s reverse, radius 0.15 m) approach head-on in an 8 m, 0.65 m-wide single-lane corridor with a pull-in bay and end-room refuges. On conflict detection each robot broadcasts a frozen *snapshot*: position, heading, speed, goal, corridor entry time, urgency, and payload state. A policy receives the two snapshots and returns a *yielder*, a *proceeder*, and a refuge; the yielder reverses into or toward the refuge, waits, then continues. The 2D unicycle simulator runs Monte Carlo encounters; we use the single-bay and two-bays layouts with 100 encounters per layout (200 total) from seeds 100000–100099. These seeds are disjoint from those used to train and evaluate the learned negotiator in [1]. Arms run at 25 encounters per layout use seeds 100000–100024, a strict subset.

An episode succeeds if both robots reach their goals without collision within the 180 s simulation horizon; if either robot has not arrived by then, the episode is a *timeout* (labeled `deadlock` in the released data). Metrics are success, collision, timeout, urgency-weighted excess delay against the best forced yielder, and *right-yielder rate*. The forced-yielder reference replays both yield assignments, each with the planner's default refuge, and keeps the better. Right-yielder rate is the share of encounters whose first decision names the reference's yielder, excluding encounters where the two forced outcomes differ by less than 0.5 s. Excess is averaged over successful episodes; success is reported separately. Whenever the LLM answer is unusable (server error, parse failure, a yielder with no feasible refuge, or disagreement in agents mode), the companion benchmark's default rulebook decides, and the fallback fraction reports how often. The fallback decision takes effect only after the charged decision time (Sec. III-B) has elapsed. Under measured latency a slow failed call can therefore push an episode past the horizon even though the rulebook supplies a decision. Every zero-cost run in this paper succeeds.

### B. LLM policy adapter

The `llm` policy issues OpenAI-compatible chat requests at decision time with temperature 0, seed 0, and a 4096-token output cap. In *mediator* mode one call describes both snapshots and asks for the yielder. In *agents* mode each robot is prompted separately ("You are robot $X$") and answers yield or proceed; exactly one yield and one proceed are required, and any other combination falls back. Two information levels exist. *Raw* gives the natural-language state: positions, headings, speeds, urgency, load, and feasible refuges with distances. *Plans* adds, for each refuge, both robots' delays predicted by the kinematic planner, the same numbers cost bidding uses. The rulebook does not see these numbers when choosing the yielder; it uses the planner only to pick the yielder's refuge. Appendix A gives the prompts verbatim.

Responses are constrained by a JSON schema with three required fields: a free-text `reason` of at most 300 characters; the decision (`yielder`, enumerated over the two robot IDs, or `action`, enumerated over yield and proceed); and a `refuge`, enumerated over the layout's refuge IDs plus `any`. A named refuge that is infeasible for the yielder is replaced by the planner's choice.

Replies are cached on a SHA-256 hash of the full request. Within a run, identical prompts therefore return identical answers, and an interrupted run resumes exactly. Caching does not make the model deterministic across runs: each run starts a fresh cache, and batched GPU inference is not bitwise reproducible at temperature 0. Rerunning one configuration in separate runs moved nemotron-omni's egocentric-agent disagreement count between 167 and 173 of 200, and thinking-off qwen3.8-27B mediator+plans excess from 11.1 to 14.9 s. We treat differences of this size as run-to-run variation.

Two cost modes isolate quality from deployment. `think=0` charges no decision time and measures decision quality alone. `think=measured` charges the real response latency: each robot evaluates the prompts itself and brakes until the answer arrives. The second robot's identical request hits the cache and is charged the original latency, so both robots stand still for the same time. In agents mode the two per-robot prompts would run on separate robots in parallel, so the charge is the slower of the two calls. Measured latency depends on how requests reach the server. Our vLLM deployment processes one sequence at a time, so runs with four concurrent episodes queue requests, and their latency includes waiting time. We label each measured arm *serial* (one episode at a time) or *concurrent* (four).

### C. Vision arms

Two renderers feed VLMs the encounter as an image. *Global*: a top-down map showing walls, refuge parking marks,

both robots as labeled discs with heading arrows, urgency/load annotations, and goal stars. *Egocentric*: the same render clipped to each robot's forward view ($-1.5$ to $+6.5$ m along its heading) — the agents mode's onboard-camera analogue, so each robot sees only what it would sense, while the broadcast text still carries the peer's state. Modality arms compare text-only, image-only, and text+image on the global view, and text vs. egocentric image on the decentralized mode.

### D. Models

We screen nine models (Table I): eight served locally on one NVIDIA GB10 node (vLLM, NVFP4 quantization, per-family reasoning parsers) plus hosted `nemotron-3-super-120b`. A tenth local model, Muse-Glimmer-30B, failed to load because the serving stack does not support its architecture. Reasoning models emit a trace before answering; a per-request switch (`enable_thinking=false`) disables it where supported. *Thinking-off* means the model emits the decision directly, and we verified that the recorded reasoning field is empty in every thinking-off episode. *Thinking-on* means it writes a trace first.

### E. Protocol

The screen runs mediator+plans with `think=0` and each model's default reasoning setting on 200 encounters. The screen rows predate trace logging; reruns of the same configuration on 50 encounters (`on0`) recorded a trace in every episode. The top three screen models then enter a matrix over mode, information level, thinking, and cost mode. Thinking-off cells run 200 encounters; thinking-on cells beyond the screen run 50, because their traces made them several times slower. A 50-encounter arm is paired with a 200-encounter arm only on the 50 shared encounters. On that subset the baselines differ (rulebook 3.7 s, fixed priority 10.3 s, versus 2.9 and 10.7 s on 200), and we report each arm against baselines on its own encounters. Vision, egocentric, and tiebreak arms run thinking-off, `think=0`, 200 encounters. Table II lists $n$, excess with 95% CI, right-yielder rate, and fallback composition for every arm we report. Per-arm CIs use the normal approximation.

We exclude several released runs. The hosted endpoint returned errors on 58–100% of calls in every cell except the screen. Runs `qwen3.8-27b-offm`, `qwen3.8-27b-c2/c4`, and `qwen3-30b-c1-c4` logged server connection resets and fell back on 64–100% of encounters in every text arm. The concurrent thinking-off measured run `qwen3-30b-offm` is used only to quantify queueing (Sec. IV-C). Fallback composition, per-layout splits, and per-episode answers (including reasoning traces where recorded) are in the released data.[1]

## IV. Results

### A. Model screen

Figure 1 and Table II screen all models on the mediator+plans arm (`think=0`, 200 encounters). Reasoning-enabled mixture-of-experts models cluster tightly: qwen3.8-27B, qwen3-30B, and hosted nemotron-3-super reach $-1.7$ s mean excess, nemotron-nano $-1.5$ s, and nemotron-omni $-1.0$ s, with right-yielder rates of 92–93%. Excess below the reference is possible because the reference fixes each yielder's refuge to the planner's choice, which minimizes *unweighted* predicted delay, whereas the prompt asks for urgency-weighted delay and the models sometimes name a different refuge. Cost bidding, which uses the same predicted delays, matches the reference (0.0 s, 99%). Nemotron-lightning reaches 1.7 s (80%), but 66% of its encounters fall back on parse failures because its traces exhaust the 4096-token cap, so its excess is largely the rulebook's. The remaining models score worse than the rulebook (2.9 s, 72%): gemma4-26B at 6.1 s (75%), nemotron-super-49B at 15.8 s (42%), and nemotron-vl-12B at 21.7 s (38%), the last two beyond fixed priority (10.7 s). A 1.5B model on a different serving stack (llama.cpp, 4-bit GGUF; 20 encounters, `think=0`) reached 28.0–30.5 s and 25–30% right in mediator mode; these runs used a 256-token output cap instead of 4096.

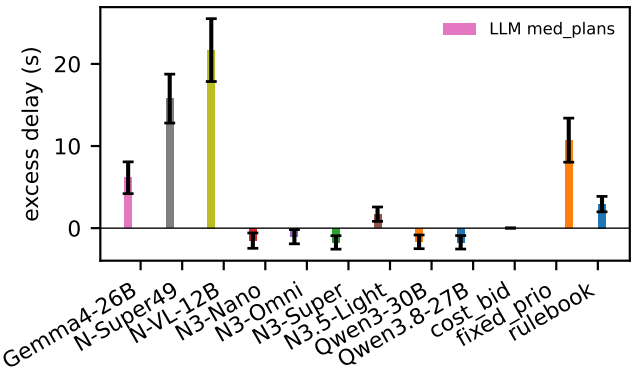



Fig. 1. Model screen, mediator+plans, `think=0`, default reasoning setting (200 paired encounters). Negative excess beats the best-forced-yielder reference. N3.5-Lightning's bar is mostly rulebook fallback (66% parse failures).

TABLE I
MODELS SCREENED (LOCAL VLLM ON ONE GB10 NODE + HOSTED ENDPOINT).

| Model | Params | Notes |
|---|---|---|
| Nemotron-3.5-Lightning | 30B-A3B | reasoning, MTP spec. |
| Nemotron-3-Nano | 30B-A3B | reasoning |
| Nemotron-3-Nano-Omni | 30B-A3B | reasoning, VL |
| Llama-3.3-Nemotron-Super | 49B | dense, reasoning |
| Nemotron-Nano-VL | 12B | VL |
| Qwen3-30B-A3B | 30B-A3B | reasoning |
| Qwen3.8-27B | 27B | reasoning, VL |
| Gemma4-26B-A4B | 26B-A4B | VL |
| nemotron-3-super-120b (hosted) | 120B-A12B | NVIDIA API |

### B. The reasoning trace carries the capability

To test whether the reasoning trace carries the capability, we rerun the top screen models with `enable_thinking=false`. On the same 200 encounters as the screen, qwen3.8-27B mediator+plans goes from $-1.7$ s (93% right) to $+11.1$ s (66%), and qwen3-30B from $-1.7$ s

[1] https://github.com/briandeegan82/multi-agent-robots (directory `amr_negotiation`).

TABLE II
RESULTS BY ARM, MEDIATOR MODE. EXCESS: MEAN ± 95% CI (S) OVER SUCCESSFUL EPISODES. RIGHT: RIGHT-YIELDER RATE. FALLBACK: SHARE DECIDED BY THE RULEBOOK, WITH DOMINANT CAUSE. $n$=50 ARMS USE SEEDS 100000–100024 PER LAYOUT.

| Model | $n$ | Excess | Right | Fallback |
|---|---|---|---|---|
| *Baselines* | | | | |
| Cost bidding | 200 | $0.0 \pm 0.0$ | 99% | – |
| Rulebook | 200 | $2.9 \pm 0.9$ | 72% | – |
| Fixed priority | 200 | $10.7 \pm 2.7$ | 56% | – |
| Rulebook | 50 | $3.7 \pm 2.1$ | 72% | – |
| Fixed priority | 50 | $10.3 \pm 5.2$ | 56% | – |
| *Screen: plans, `think=0`, default reasoning* | | | | |
| Qwen3.8-27B | 200 | $-1.7 \pm 0.8$ | 93% | 0% |
| Qwen3-30B | 200 | $-1.7 \pm 0.8$ | 92% | 0% |
| N3-Super (hosted) | 200 | $-1.7 \pm 0.8$ | 93% | 0.5% error |
| N3-Nano | 200 | $-1.5 \pm 0.9$ | 92% | 0% |
| N3-Omni | 200 | $-1.0 \pm 0.9$ | 93% | 0.5% parse |
| N3.5-Lightning | 200 | $1.7 \pm 0.9$ | 80% | 66% parse |
| Gemma4-26B | 200 | $6.1 \pm 1.9$ | 75% | 0% |
| N-Super-49B | 200 | $15.8 \pm 3.0$ | 42% | 0% |
| N-VL-12B | 200 | $21.7 \pm 3.8$ | 38% | 0.5% parse |

| Model | Condition | $n$ | Excess | Right | Fallback |
|---|---|---|---|---|---|
| *Reasoning off, `think=0`* | | | | | |
| Qwen3.8-27B | plans | 200 | $11.1 \pm 2.5$ | 66% | 0% |
| Qwen3.8-27B | raw | 200 | $23.2 \pm 4.7$ | 40% | 0% |
| Qwen3-30B | plans | 200 | $22.5 \pm 3.7$ | 35% | 0% |
| Qwen3-30B | raw | 200 | $29.9 \pm 5.6$ | 36% | 0% |
| *Reasoning on, `think=0`, 50-encounter subset* | | | | | |
| Qwen3.8-27B | plans | 50 | $-1.3 \pm 1.4$ | 94% | 0% |
| Qwen3.8-27B | raw | 50 | $3.7 \pm 2.1$ | 72% | 98% error |
| Qwen3-30B | plans | 50 | $-1.3 \pm 1.4$ | 94% | 0% |
| Qwen3-30B | raw | 50 | $3.7 \pm 2.2$ | 74% | 80% parse |
| *Measured latency (regime, mean charged latency)* | | | | | |
| Qwen3-30B | plans, on, conc., 31.1 s | 50 | $103.1 \pm 15.3$ | 92% | 0% |
| Qwen3-30B | raw, on, conc., 100.3 s[a] | 50 | $301.5 \pm 42.7$ | 72% | 88% parse |
| Qwen3-30B | plans, off, serial, 0.7 s | 200 | $23.6 \pm 3.5$ | 37% | 0% |
| Qwen3.8-27B | plans, off, serial, 6.7 s | 200 | $34.1 \pm 3.4$ | 64% | 0% |

[a] 64% success; excess over successful episodes only.

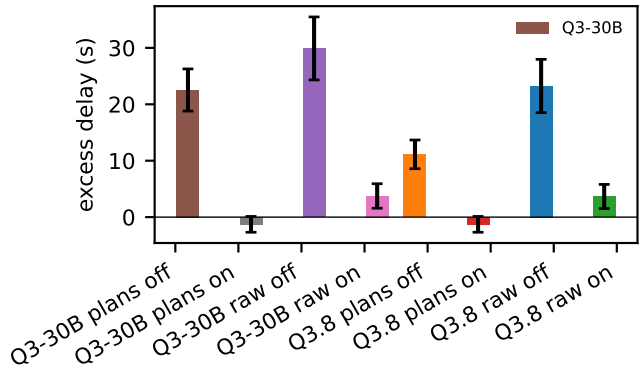



Fig. 2. Reasoning on vs off, mediator arms, `think=0`. Off cells: $n$=200; on cells: $n$=50 (first 25 seeds per layout). The raw-on bars are mostly rulebook fallback (qwen3.8 98% errors, qwen3-30B 80% parse failures) and do not measure reasoning-on quality.

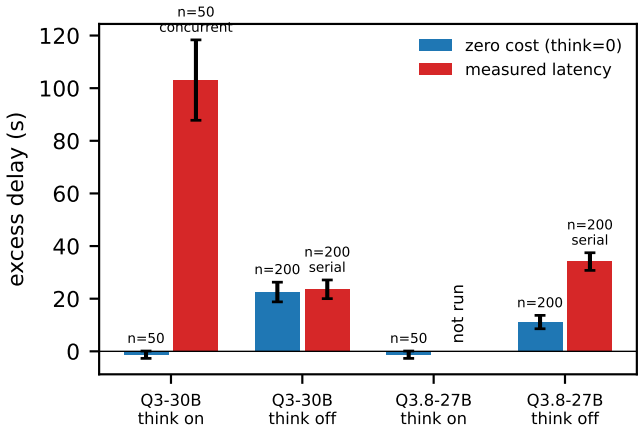



Fig. 3. Zero-cost vs measured-latency excess, mediator+plans. Each pair shares encounters ($n$ above bars). The thinking-on qwen3-30B measured arm ran concurrently, so its latency includes queueing. No completed measured thinking-on arm exists for qwen3.8 (marked; a serial run was stopped after 10 of 50 encounters).

(92%) to +22.5 s (35%) (Table II, Fig. 2). On raw prompts, thinking-off gives +23.2 s (40%) and +29.9 s (36%). Fig. 2 plots the thinking-on cells from the 50-encounter reruns (−1.3 s for both models) rather than the screen. Paired per-encounter differences for the plans arms are [XX] s and [XX] s (95% CI). The third top model, hosted nemotron-3-super, returned errors on 97% of thinking-off calls and is excluded. Thinking-off models receive the same planner delays that let cost bidding match the reference; without the trace, they do not turn those numbers into the right choice. The thinking-on raw cells cannot serve as a comparison. On their 50 encounters, 98% (qwen3.8, server errors) and 80% (qwen3-30B, parse failures) of decisions fell back to the rulebook, so those bars in Fig. 2 show the rulebook's excess.

### C. The deployed cost

Under `think=measured` both robots stand still until the answer arrives (Fig. 3, Table II). For thinking-on qwen3-30B mediator+plans (50 encounters, concurrent), excess rises from −1.3 s at zero cost on the same encounters to +103.1 s at a mean charged latency of 31.1 s, while the right-yielder rate is 92%, against 94% at zero cost. The slope follows from the cost model rather than a fit: two robots idling at the mean urgency weight of 5/3 accrue about 3.3 s of weighted delay per second of latency. Both measured arms corroborate it: the 104.4 s increase over 31.1 s is 3.4 s/s, and serial thinking-off qwen3.8-27B goes from 11.1 to 34.1 s at 6.7 s latency, also 3.4 s/s. The serial thinking-off qwen3-30B arm (0.7 s) is too fast to resolve the slope against its ±3.5 s CI. The decentralized measured arms in the same runs fare worse (thinking-on qwen3-30B agents+plans, concurrent: 172.9 s excess at 51.9 s latency) and are in the released data.

The thinking-on measurement is an upper bound on what a single deployed mediator would pay. Its arm ran concurrently, and queueing dominates concurrent latency: the same model with thinking off measured 22.0 s mean latency concurrently and 0.7 s serially. A serial thinking-on rerun (qwen3.8-27B)

was stopped after 10 of 50 encounters had taken 9.8 h. A serial thinking-on run of the hosted nemotron-3-super completed, but every call returned an error within 0.1 s, so it measures error latency, not deliberation. No informative serial thinking-on measurement is available. On raw prompts, thinking-on with measured latency succeeds in only 64% of 50 encounters. There, 88% of decisions are parse failures after traces hit the token cap, the mean charged latency is 100.3 s, and the rulebook decision that follows arrives too late for both robots to finish within the 180 s horizon. The 301.5 s excess over the successful 32 episodes excludes the slowest cases, so we do not compare it with other arms.

The slope gives a break-even estimate. On the 200-encounter screen, a reasoning mediator has a quality margin of 1.0–1.7 s against cost bidding, which survives only if mean latency stays below about 0.3–0.5 s. Against the rulebook (2.9 s) the margin is 3.9–4.6 s, and the latency limit is about 1.1–1.4 s. Thinking-off models reach sub-second latency (0.7 s for qwen3-30B) but lose the margin itself (23.6 s excess). No measured configuration combined sub-second latency with reasoning-level quality. At current inference speeds, cost bidding, which computes its bids from the planner without model inference, dominates; an LLM negotiator becomes viable only if deliberation leaves the critical path through precomputation, asynchronous negotiation, or much faster inference.

### D. Vision input

Four VL-capable models ran three mediator arms with thinking off, `think=0`, 200 encounters (Fig. 4): text with planner delays (plans), the global image alone with no text, and the image with raw text (no planner delays). The image alone gives lower excess than text+plans on all four models: gemma4 3.9 vs 6.6 s, qwen3.8 9.0 vs 14.9 s, nemotron-omni 14.2 vs 16.4 s, and vl-12b 17.0 vs 22.1 s. The per-arm 95% CIs overlap for all four models (e.g., omni $\pm$3.7 and $\pm$3.1 s), so this ordering is not established until paired differences are computed: [XX]. Adding raw text to the image makes every model worse than the image alone: gemma4 6.4, qwen3.8 13.2, omni 20.6, and vl-12b 30.0 s (per-arm CIs overlap for gemma4 and qwen3.8). Because the text in that arm lacks the planner delays, we cannot separate interference between modalities from the change in information. All image arms remain worse than reasoning-on text ($-1.0$ to $-1.7$ s): with reasoning off, the image recovers part of the lost quality but not all of it.

### E. Egocentric decentralized agents

Chance encounters between unaffiliated robots motivate the agents mode: each robot gets only its egocentric view plus the peer's broadcast and decides independently. All arms here run thinking off, `think=0`, 200 encounters. Disagreement, not decision quality, dominates, and its form differs by model (Fig. 5). Nemotron-omni has both robots volunteer to yield on 84% of encounters (167 of 200) and never has both proceed on this arm; the rulebook then decides. Qwen3.8-27B and vl-12b have both robots proceed on 29% and 31%; each robot's reason cites proximity facts that assign the yield to the other robot. Gemma4 agrees on 55%, and 85 of its 91 conflicts

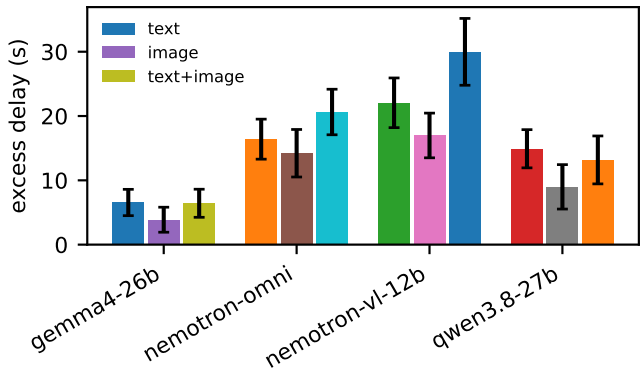



Fig. 4. Modality arms, mediator, thinking off, $n$=200. The text arm includes planner delays; text+image uses raw text. Image-only has the lowest mean excess on all four VL models, but its per-arm CI overlaps the text arm's for each.

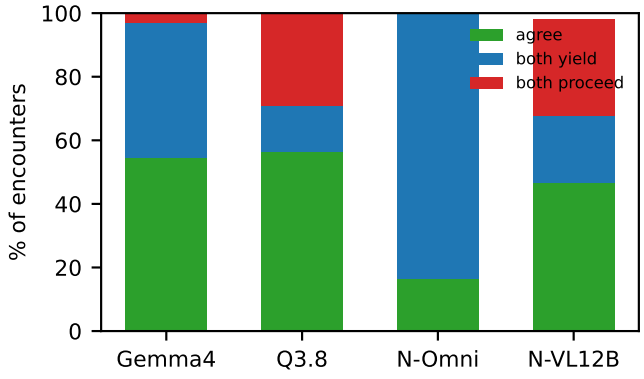



Fig. 5. Action-agreement modes, egocentric agents arm, thinking off, $n$=200 per model. Conflicts are mostly both-yield for omni and gemma4 and mostly both-proceed for qwen3.8 and vl-12b.

are both-yield. Agreement does not imply a good choice. Excluding ties, the right-yielder rate on agreed encounters is 58% for omni (19 of 33), 69% for qwen3.8 (77 of 112), 64% for vl-12b (59 of 92), and 71% for gemma4 (77 of 108). Wrong-yielder agreements (14, 35, 33, and 31 encounters) execute without any fallback, at a mean excess of [XX] s. Ego images give lower excess than raw text for vl-12b (5.3 vs 7.4 s) and qwen3.8 (8.9 vs 10.0 s) but higher for omni (3.4 vs 3.0 s) and gemma4 (3.8 vs 2.8 s). Omni's raw-text arm falls back on 97% of encounters, so its excess is essentially the rulebook's. Planner delays help only gemma4 agree: with plans it disagrees on 21% of encounters and reaches 88% right-yielder at 3.1 s, whereas qwen3.8, omni, and vl-12b disagree on 137, 199, and 194 of 200, more than with egocentric images (87, 167, and 103). No input level gave all four models a shared convention.

### F. The protocol default

Disagreement calls for a tiebreak. We test a random draw in place of the rulebook fallback. Each robot's card is the SHA-256 hash of its ID, both IDs, and the encounter's detection time; both robots compute it from broadcast data, neither

TABLE III
PROTOCOL-DEFAULT ARMS, EGOCENTRIC AGENTS, THINKING OFF, $n$=200: MEAN EXCESS IN S (RIGHT-YIELDER RATE). RULEBOOK AND HIDDEN DRAW COME FROM THE SAME RUN AND SHARE ANSWERS; THE VISIBLE DRAW IS A SEPARATE RUN. PER-ARM 95% CIS: ±0.9–3.4 S.

| Model | Rulebook[a] | Hidden draw | Visible draw |
|---|---|---|---|
| N3-Omni | 3.1 (71%) | 13.1 (49%) | 15.1 (48%) |
| Qwen3.8-27B | 10.6 (64%) | 14.0 (57%) | 12.2 (57%) |
| Gemma4-26B | 4.1 (71%) | 7.8 (63%) | 7.6 (59%) |
| N-VL-12B | 5.4 (63%)[b] | 10.0 (53%)[b] | 10.5 (58%) |

[a]Paired with the hidden draw. The visible-draw runs contain their own rulebook arm: 2.7 (73%), 8.7 (62%), 2.5 (75%), and 4.8 (68%), top to bottom.
[b]10 encounters (5%) fall back on parse failures.

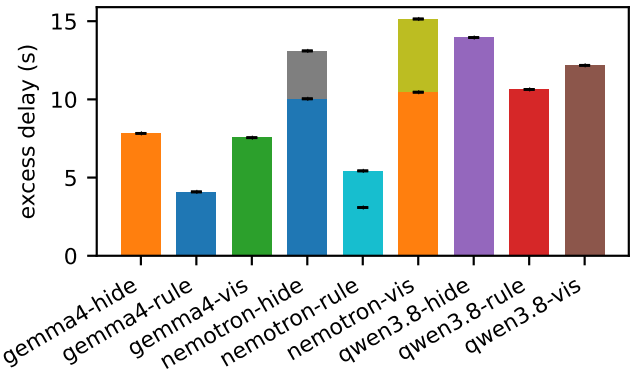



Fig. 6. Protocol-default arms: rulebook fallback vs hidden vs visible random tiebreak (egocentric agents, thinking off, $n$=200, same encounters).

chooses its value, and the larger card proceeds on conflict. The draw is fair only between non-adversarial robots, because a robot could shift the detection time it broadcasts; a commit-reveal protocol [13] would remove that lever, and we did not implement one. Across the 200 encounters the larger card belongs to robot A in [XX]% of encounters. Three arms, all egocentric agents with thinking off, separate presentation from resolution (Table III, Fig. 6): the rulebook default; a hidden draw, whose prompts are byte-identical to the rulebook arm; and a visible draw, whose prompt states the rule and both card values (Appendix A).

The hidden draw isolates resolution. Its prompts match the rulebook arm, so within a run its answers are the same cached responses (omni: 170 both-yield, 27 agreed, 3 both-proceed in both arms), and only the resolution of conflicts changes. On the conflicted encounters the rulebook names the right yielder 66–71% of the time and the draw 40–51%: omni 71 vs 47%, qwen3.8 67 vs 51%, gemma4 70 vs 51%, and vl-12b 66 vs 40%. Arm-level excess rises from 3.1–10.6 s to 7.8–14.0 s, and right-yielder rates fall from 63–71% to 49–63%.

The visible draw adds presentation. The draw still resolves every conflict, so no image-arm encounter falls back to the rulebook (vl-12b's text-only arm has 10 parse fallbacks), but conflicting answers persist. Both robots yield or both proceed on 126 (omni), 83 (qwen3.8), 102 (gemma4), and 104 (vl-12b) of 200 encounters, against 169, 80, 84, and 97 under the rulebook default in the same runs. Only omni engages with the draw at scale. At least one omni robot cites the card or draw on 183 of 200 encounters ("A's card is smaller than B's, so if both yield or proceed, B would proceed"), against 28, 27, and 2 for the other three models, and only omni's conflict count falls. Outcomes match the hidden draw: right-yielder 48–59% and excess 7.6–15.1 s with egocentric images, and 45–56% and 8.2–16.6 s with text only.

A fair coin is therefore a poor default here: it resolves every conflict but discards the information the rulebook uses on exactly the encounters where agents fail to agree. Showing the coin changed what one model wrote, not how often the right robot yielded.

## V. DISCUSSION

Decision quality and deployment cost point in different directions. Where a shared decision service is acceptable, a reasoning mediator with planner delays already scores at or below the forced-yielder reference at zero decision cost. The engineering problem is to take deliberation off the critical path, for example by negotiating predicted encounters ahead of time, caching likely decisions, or routing routine cases to a fast tier. For chance encounters between unaffiliated robots, decentralized foundation-model agents are not yet deployable: in the egocentric arm, conflicting answers occur on 44–84% of encounters, depending on model, and the fallback then decides. The fallback therefore matters. An informed rulebook recovers 66–71% right-yielder decisions on those encounters, whereas a fair coin recovers 40–51%, whether or not the agents see it.

### A. Limitations

All results are from simulation; the Gazebo transfer validated in [1] covers classical policies, not the LLM arms. The benchmark, baselines, and reference come from the companion submission [1], which is under review; Sec. III restates the elements this paper relies on. Two layouts and one encounter type bound the generality. The measured thinking-on result comes from one model on 50 encounters under concurrent execution, and no informative serial thinking-on measurement exists. Thinking-on cells beyond the screen use 50 encounters. All local models ran with NVFP4 quantization; absolute decision quality may differ at full precision. Runs of identical configurations vary (Sec. III-B), and we report one run per arm. The hosted endpoint failed on 58–100% of calls outside the screen. Thinking-off is honored on the models tested but is not universal across serving stacks, and several models' traces exceed the 4096-token cap on raw prompts, as the parse-fallback counts show. Agents answer a single-shot prompt; multi-turn dialogue, tool use (querying the planner), and selfish-utility prompts are untested.

## VI. CONCLUSION

Reasoning-enabled language models can negotiate head-on robot encounters at or below the forced-yielder reference when decision time is free, and the reasoning trace, not the model weights alone, carries that ability: with the trace disabled, the

same models fall below a hand-written rulebook. Deployed viability depends on latency. Each second of latency costs about 3.4 s of excess, so at measured inference speeds cost bidding remains the better mechanism. For decentralized agents, the bottleneck is coordination, not perception: models disagree in model-specific ways that richer inputs do not remove for all of them. When agents disagree, the protocol's fallback decides the outcome, and an informed fallback outperforms a fair coin.

## Appendix A
## Prompt templates

The prompts below are copied from `amr_negotiation/llm.py`. Angle brackets mark fields filled from the snapshots, a vertical bar separates alternatives, and bold or italic labels are not part of the prompt. Situation lines are joined by newlines; the agent prompt places a blank line before its question.

**System (all modes).** You coordinate autonomous mobile robots (TurtleBot3 Waffle) in a warehouse. The corridor is a single lane: two robots cannot pass each other inside it. One robot must pull into a refuge (a side bay or an end room) and wait while the other passes. Reversing is slower than driving forward. Answer only with JSON.

**Situation.** Corridor: single lane from x=0 to x=⟨L⟩ m, ⟨W⟩ m wide. Time t=⟨t⟩ s.
*Per robot:* robot ⟨id⟩: x=⟨x⟩ m, inside the corridor | outside the corridor (in an end room), heading +x (east) | −x (west), goal x=⟨g⟩ m, speed now ⟨v⟩ m/s (max forward ⟨$v_f$⟩, max reverse ⟨$v_r$⟩ m/s), urgency ⟨u⟩ (1 = normal; higher = its delay costs more), loaded (carrying a payload) | unloaded, entered the corridor at t=⟨$t_e$⟩ s | not yet in the corridor
*Per robot Y, other robot P:* Refuges robot ⟨Y⟩ could pull into to let ⟨P⟩ pass: | Robot ⟨Y⟩ has no refuge it can pull into.
*Per feasible refuge:* ⟨id⟩: ⟨bay|room⟩ at x=⟨x⟩ m, ⟨d⟩ m behind it (it must reverse) | ahead of it
*Plans only, appended:* ; predicted delay if ⟨Y⟩ yields here: ⟨Y⟩ ⟨$d_Y$⟩ s, ⟨P⟩ ⟨$d_P$⟩ s

**Mediator.** ⟨Situation⟩ Decide which robot yields so that the urgency-weighted total delay of both robots is as small as possible. Reply as JSON: {"reason": "<one sentence>", "yielder": "<robot id>", "refuge": "<refuge id, or any>"}

**Agent.** You are robot ⟨me⟩. Robot ⟨other⟩ has broadcast its state to you. ⟨Situation⟩ Should you (⟨me⟩) yield, or proceed and let ⟨other⟩ yield? Aim to keep the urgency-weighted total delay of both robots small; if you both proceed or both yield, a fixed rulebook decides instead. Reply as JSON: {"reason": "<one sentence>", "action": "yield" or "proceed", "refuge": "<refuge id, or any>"}

**Visible draw.** "a fixed rulebook decides instead." becomes "the draw decides instead.", and before the JSON instruction: Draw rule: each robot holds a card | a random value bound to its id and this encounter that neither can choose: yours is ⟨$c_{me}$⟩, ⟨other⟩'s is ⟨$c_{other}$⟩. If actions conflict (both yield or both proceed), the robot with the LARGER card proceeds.

**Image legends (appended to the text).** *Global:* The image is a top-down map of the same encounter: the corridor and its refuges (x marks are parking poses), each robot is a coloured disc (the first robot listed is blue, the second is orange) with a black arrow showing its heading, and the matching star is its goal. Numbers in the text are authoritative if they disagree. *Egocentric:* The image is your own forward view from your onboard camera: the corridor and its refuges (x marks are parking poses), you are the blue disc with a black heading arrow, the star is your goal, and the orange disc is the other robot if it is in view.

In image-only mediator arms the situation text is omitted. Hidden-draw prompts are identical to the rulebook-arm agent prompt.

## References


[1] Anonymous, "Who backs into the bay? benchmarking negotiation policies for head-on robot encounters in single-lane corridors," 2026, companion submission.

[2] T. Godfrey, W. Hunt, and M. D. Soorati, "MARLIN: Multi-Agent Reinforcement Learning Guided by Language-Based Inter-Robot Negotiation," arXiv preprint, 2024. [Online]. Available: https://arxiv.org/abs/2410.14383

[3] V. Mahadevan, S. Zhang, and R. Chandra, "GameChat: Multi-LLM Dialogue for Safe, Agile, and Socially Optimal Multi-Agent Navigation in Constrained Environments," in *2025 IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, 2025, pp. 1–7. [Online]. Available: https://doi.org/10.1109/mrs66243.2025.11357267

[4] K. Garg, S. Zhang, J. Arkin, and C. Fan, "Foundation Models to the Rescue: Deadlock Resolution in Connected Multi-Robot Systems," arXiv preprint, 2024. [Online]. Available: https://arxiv.org/abs/2404.06413

[5] Y. Chen, J. Arkin, Y. Zhang, N. Roy, and C. Fan, "Scalable Multi-Robot Collaboration with Large Language Models: Centralized or Decentralized Systems?" in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 4311–4317. [Online]. Available: https://doi.org/10.1109/icra57147.2024.10610676

[6] M. Dias, R. Zlot, N. Kalra, and A. Stentz, "Market-Based Multirobot Coordination: A Survey and Analysis," *Proceedings of the IEEE*, vol. 94, no. 7, pp. 1257–1270, 2006. [Online]. Available: https://doi.org/10.1109/jproc.2006.876939

[7] B. Gerkey and M. Mataric, "Sold!: auction methods for multirobot coordination," *IEEE Transactions on Robotics and Automation*, vol. 18, no. 5, pp. 758–768, 2002. [Online]. Available: https://doi.org/10.1109/tra.2002.803462

[8] O. Amir, G. Sharon, and R. Stern, "Multi-Agent Pathfinding as a Combinatorial Auction," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 29, no. 1, 2015. [Online]. Available: https://doi.org/10.1609/aaai.v29i1.9427

[9] A. Gautier, A. Stephens, B. Lacerda, N. Hawes, and M. Wooldridge, "Negotiated Path Planning for Non-Cooperative Multi-Robot Systems," in *International Joint Conference on Autonomous Agents and Multiagent Systems*, 2022, pp. 472–480. [Online]. Available: https://doi.org/10.65109/dvsq3214

[10] N. K. Alghamdi and S. Park, "Opinion-Driven Decision-Making for Multi-Robot Navigation through Narrow Corridors," arXiv preprint, 2025. [Online]. Available: https://arxiv.org/abs/2504.20947

[11] R. J. Aumann, "Subjectivity and correlation in randomized strategies," *Journal of Mathematical Economics*, vol. 1, no. 1, pp. 67–96, 1974.

[12] J. F. Nash, "The Bargaining Problem," *Econometrica*, vol. 18, no. 2, p. 155, 1950. [Online]. Available: https://doi.org/10.2307/1907266

[13] M. Blum, "Coin flipping by telephone: A protocol for solving impossible problems," *SIGACT News*, vol. 15, no. 1, pp. 23–27, 1983.